

A Functional Regression Framework for Genetic Colocalization analysis across Omics and Disease-related Outcomes

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What is colocalization?

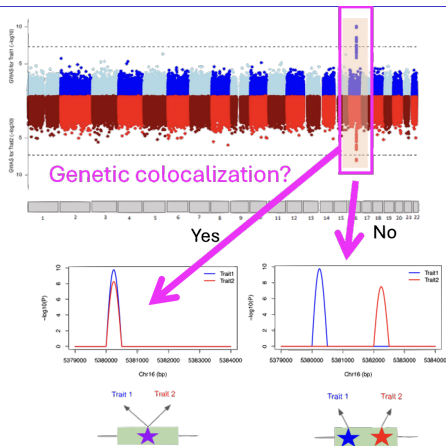
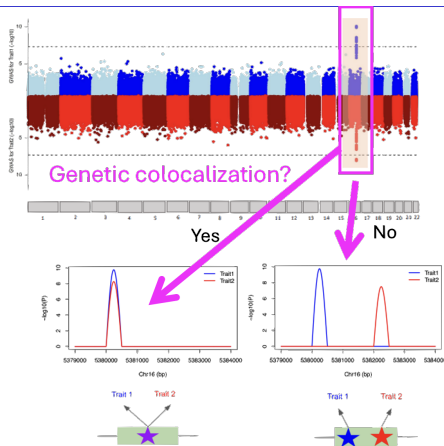


Figure adapted from Zang et al, 2024, Osteoarthritis and Cartilage Open

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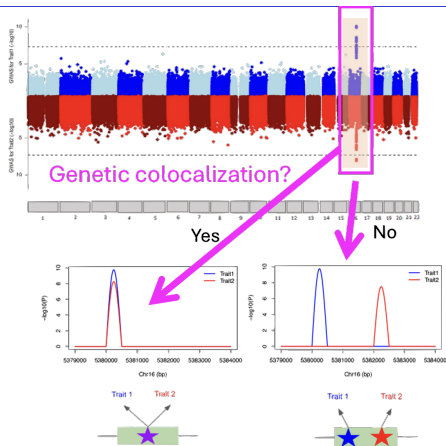


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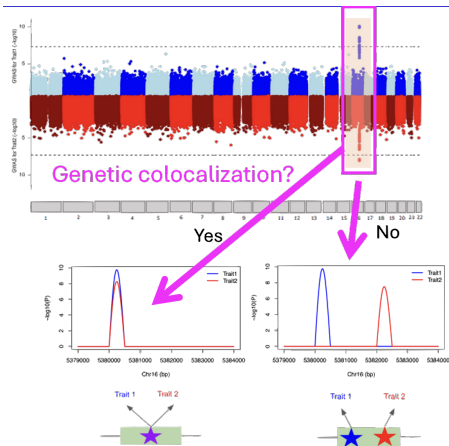


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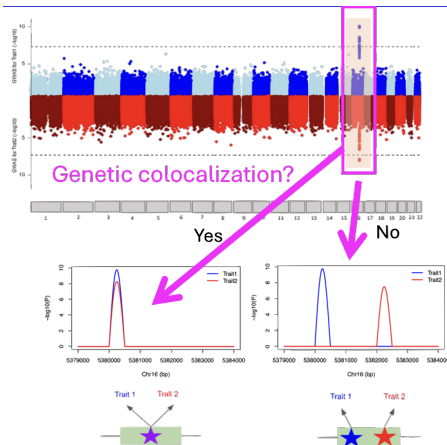


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- MicroRNAs (miRNAs) play an important role in regulating gene expression and are increasingly reported as associated with OA outcomes.

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- Colocalization methods to combined analysis of OA outcomes and miRNA levels can shed light on potential regulatory mechanisms.
- Increasing availability of individual-level multi-omic data.

Objective: To develop a multi-trait multi-variant colocalization approach with mixed trait distributions using individual-level data.

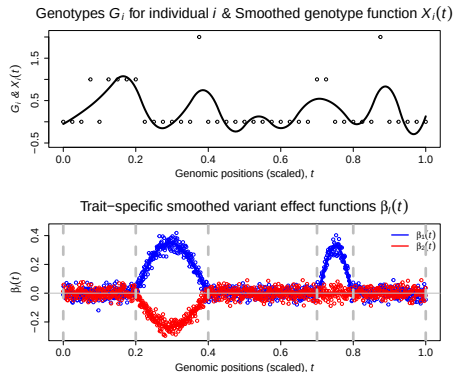
FunColoc - Model Formulation

Functional regression model for colocalization analysis

- Let y_{il} be the l th trait for the i th individual and $X_i(t)$ a continuous genotype function, which are related by the following scalar-on-function regression model

$$g_l^{-1}(E[y_{il}]) = \alpha_{0l} + \int_0^1 X_i(t)\beta_l(t)dt$$

- $g_l^{-1}(\cdot)$: trait-specific inverse link function;
- α_{0l} : intercept for trait l ;
- $\beta_l(t)$: unknown continuous genetic effect function;
- $X_i(t)$ is a smoothed function based on $\mathbf{G}_i = (G_{i1}, \dots, G_{iM})'$ genotypes.



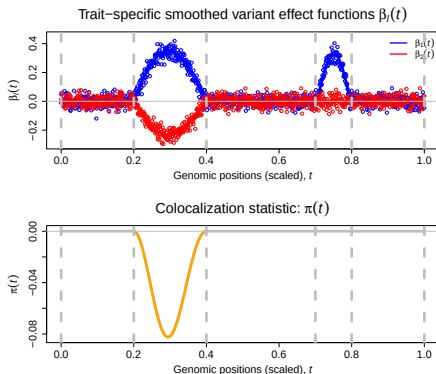
FunColoc - Model Formulation

Functional regression model for colocalization analysis

- For two traits, $l = 1, 2$ and letting $\pi(t) = \beta_1(t)\beta_2(t)$, we are interested in testing

$$H_0 : \pi(t) = 0 \text{ vs. } H_1 : \pi(t) \neq 0.$$

- Under H_0 (No colocalization)
 - $\beta_1(t) = 0$ AND $\beta_2(t) = 0$, OR
 - $\beta_1(t) = 0$ AND $\beta_2(t) \neq 0$, OR
 - $\beta_1(t) \neq 0$ AND $\beta_2(t) = 0$
- Under H_1 (Colocalization)
 - $\beta_1(t) \neq 0$ AND $\beta_2(t) \neq 0$

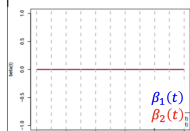


- Estimation and inference in FunColoc follows a two-step procedure using a (B-spline) basis expansion representation for $\int_0^1 X_i(t)\beta_l(t)dt$ and a multivariate covariance generalized linear model.
- Composite H_0 : *unknown* theoretical distribution
- Both local and global, i.e., locus-level, colocalization tests can be derived.
- Composite null hypothesis evaluated under various approaches: permutation and adaptive bootstrap).

Simulation Settings

Scenarios #1 to #3 under H_0 (no colocalization)

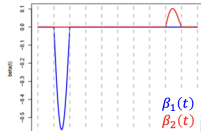
Scenario #1 (H_{00})



Scenario #2 (H_{01})

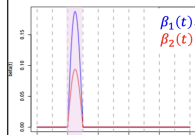


Scenario #3 (H_{01}, H_{02})

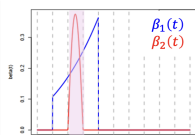


Scenarios under H_1 (colocalization)

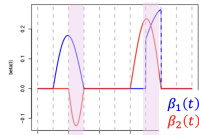
Scenario #4



Scenario #5

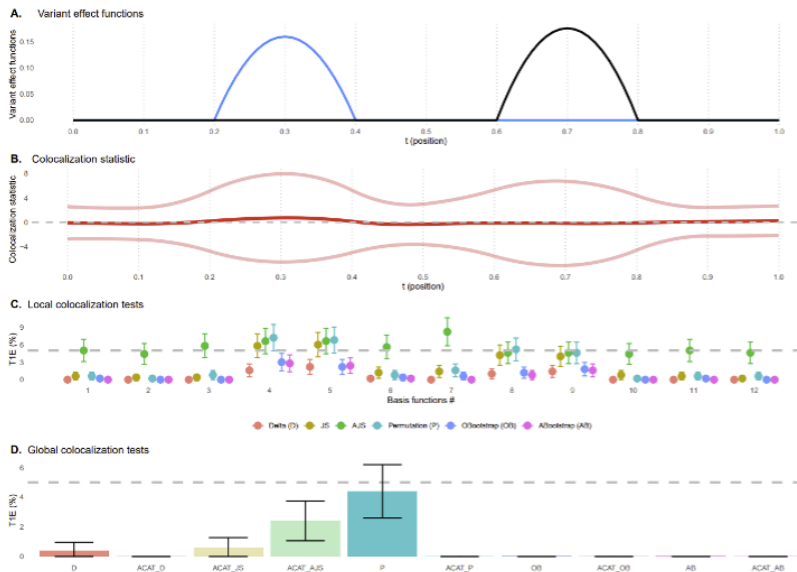


Scenario #6

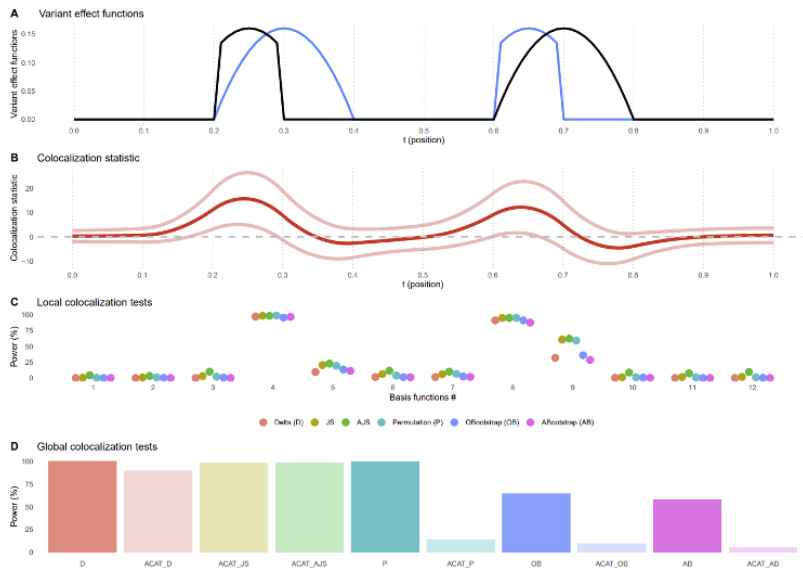


- Under H_0 :
 - Controlled ($<5\%$) in all 3 scenarios.
 - Conservative when $\beta_1(t) = 0$ AND $\beta_2(t) = 0$ (Scenario 1).
- Under H_1 :
 - Increased power when both variant effect functions fully overlap (Scenario 4).

Simulation Results - Type I error (Scenario 3)



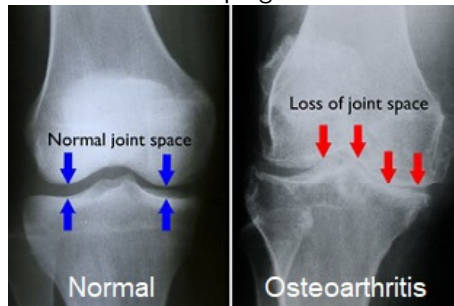
Simulation Results - Power (Scenario 6)



Application: The Osteoarthritis Initiative (OAI)

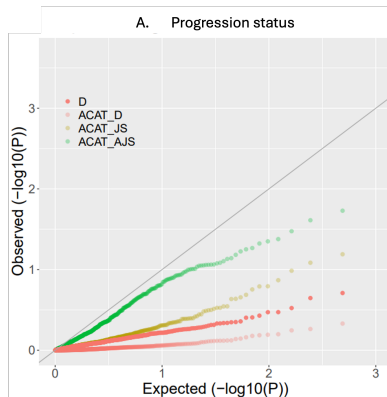
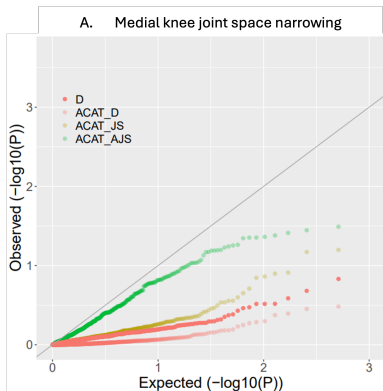
- OA is a chronic joint condition that affects 1 in 7 Canadians.
- Most commonly presented in the knee joints, with no cure nor approved disease-modifying drug therapies.
- Complex etiology with interplay between genetic, epigenetic and environmental factors.
- OAI: Multi-center, ten-year observational study of $\approx 4,800$ participants.
- Goal is to provide resources to better understand knee OA.
 $\Rightarrow$ Prevention and Treatment.

Joint Space Narrowing $\Rightarrow$ measure of knee OA progression



Application Results

$N = 303$ individuals with genotyping (Omni-Quad 2.5M), microRNA-seq (Illumina NextSeq550) & X-ray data.



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- Smoothed functions for genotypes & variant effects depend on variant positions
⇒ account for potential **non-linear genetic effects** (though parameter tuning is necessary).
- Colocalization testing:
 - FunColoc detects colocalization even in scenarios where genetic effect functions for both traits do not fully overlap
 - use of empirical approaches to account for the composite nature of H_0
 - conservative in scenarios with no genetic effects across both traits

Ongoing and Future Work

- Implement FunColoc to more than two traits.
- Extensions to function-on-function regression can be appealing.
- Integrate multi-omic data in a more strategic cost-efficient manner (e.g., two-phase studies).

Acknowledgements

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